Equivalent Expressions: Expanded & Factored Forms

expanding, factoring, and combining like terms to write equivalent expressions

Directions:

- After the warm-up the problems are mixed: read each one and decide whether to expand, factor, or combine before you start writing.
- Answers are expressions — keep the variable, and write factored answers with the greatest common factor outside.

1. Expand: $3(x+7)$

Answer: ________________

2. Expand: $5(2x+3)$

Answer: ________________

3. Factor out the greatest common factor: $4x+12$

Answer: ________________

4. Combine like terms: $8x+2-3x+5$

Answer: ________________

5. Factor out the greatest common factor: $14x + 21$

Answer: ____________________

6. Expand: $-4(x - 6)$

Answer: ____________________

7. Combine like terms: $2x + 9 + 4x - 9$

Answer: ____________________

8. Factor out the greatest common factor: $18x + 24$

Answer: ____________________

9. Consider the two expressions $2(3x+1)$ and $6x+2$.

(a) Evaluate $2(3x+1)$ when $x=4$: _________

(b) Evaluate $6x+2$ when $x=4$: _________

(c) Say whether the two expressions are equivalent for *every* value of x, and how you know.

10. A rectangular banner has area $(12x+30)$ square centimeters. Its height is 6 cm. Write and simplify an expression, in centimeters, for the width of the banner.

Answer: __________________ cm

11. Simplify completely: $4(2x+3)-5x$

Answer: __________________

12. Challenge. Consider the expression $6(x+2)+3x+6$.

(a) Write it in simplest *expanded* form.

Answer: ________________

(b) Now *factor* your answer from part (a).

Answer: ________________